Improvement of Temperature and Emissivity Separation Algorithm for Thermal Infrared Hyperspectral Imaging Based on Airborne Data

Xia Zhang, Chengyu Liu, Ruohan Chen, Biao Zeng, Haoran Xiong, Kaiyu Wang, and Suya Zhao

***Abstract*— Temperature and emissivity separation (TES) is a crucial process for converting thermal infrared (TIR) hyperspectral data into actionable information. Since the development of thermal infrared hyperspectral imagers is still in its nascent stage, most TES algorithms have been validated primarily using simulated data within the context of land resource remote sensing. However, there has been insufficient focus on urban environments with complex underlying surfaces, particularly on low emissivity targets. In this study, we developed a TES algorithm capable of adapting to a broader emissivity range. The performance of the algorithm in retrieving temperature and emissivity was evaluated against several typical TES algorithms, using data cubes acquired by the airborne thermal infrared hyperspectral imaging system (ATHIS) as test data. The experimental results revealed that existing algorithms exhibited relatively high retrieval errors for low emissivity ground objects. In contrast, the developed algorithm significantly improved the accuracy of retrieval for such objects while maintaining comparable accuracy for non-low-emissivity targets. These findings suggest that the proposed algorithm enhances TES accuracy and expands the applicability of thermal infrared hyperspectral imaging in environmental remote sensing of urban environments with complex underlying surfaces.**

***Index Terms*— Land surface emissivity, land surface temperature, low emissivity, thermal infrared (TIR) hyperspectral, urban environment.**

I. Introduction

PREVIOUS studies have demonstrated that thermal infrared (TIR) hyperspectral remote sensing offers significant advantages in various fields, including resource exploration [1], [2], [3], surface environmental monitoring [4], [5], [6], [7], [8], and air pollution detection [9], [10]. These applications, however, require precise inversion of ground object temperature and emissivity [1], [2]. One of the most challenging aspects of this process is temperature and emissivity separation (TES), which involves solving a set of underdetermined equations: N equations (representing observed radiances from N spectral bands) with $N+1$ unknowns (N band emissivities and 1 temperature). This issue has been a persistent challenge for researchers, leading to the development of numerous TES algorithms employing different strategies.

Becker and Li [11] introduced the temperature-independent spectral indices (TISI) method, which transforms the ground-leaving radiance spectrum to approximate the spectral shape of emissivity using the Planck function. The day and night algorithm based on TISI demonstrated relatively high-temperature retrieval accuracy for multiband thermal infrared remote sensing data. While TISI is theoretically applicable to thermal infrared remote sensing data with high spectral resolution, it requires additional constraints, either from data or theory, to determine specific emissivity values, as it cannot retrieve absolute emissivity values. To address this limitation, Borel [12] proposed the iterative spectral smooth temperature emissivity separation (ISSTES) method, leveraging the fact that surface emissivity spectra tend to be smoother than atmospheric downward radiance spectra. ISSTES, which represents a significant advancement in thermal infrared hyperspectral TES algorithms, establishes a cost function based on the smoothness of the emissivity spectrum to assess the residual atmospheric absorption lines in a previously retrieved emissivity spectrum. The optimal temperature, which minimizes the cost function, is determined iteratively, with the emissivity spectrum calculated according to the radiative transfer equation. In 2008, ISSTES was refined and renamed the automatic retrieval of temperature and emissivity (ARTEMISS) [13], wherein the smoothness criterion was replaced by the standard deviation between the simulated and measured sensor radiance spectra as the cost function for temperature optimization. The simulated sensor radiance spectrum is computed using the radiative transfer equation, with the temperature derived from the optimal algorithm and the corresponding smoothed emissivity spectrum. Borel et al. [14] later developed the quick temperature-emissivity separation (QTES) algorithm, based on ARTEMISS, for retrieving near-ground thermal hyperspectral data. QTES simplifies the process by focusing on a narrow spectral range (~9.5–10 μm) to estimate the optimal temperature, rather than analyzing the entire spectrum.

Building on the concept of iterative spectral smoothing, the spectral smoothness method (SpSm), along with the downwelling radiation residual index (DRRI) [15], was

Received 3 October 2024; revised 3 December 2024; accepted 16 December 2024. Date of publication 20 December 2024; date of current version 6 January 2025. This work was supported in part by the Science Research Project of Hebei Education Department under Grant QN2020212. *(Corresponding author: Chengyu Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3520865

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proposed by [16]. Simultaneously, the correlation-based TES (CBTES) method was developed, based on the theoretical assumption that the emissivity spectrum and the downward atmospheric spectrum are uncorrelated. However, it was observed that residual atmospheric absorption lines in the retrieved emissivity spectrum could increase the correlation between the emissivity and atmospheric downward spectra. As a result, the correlation coefficient between the two spectra was used as a cost function to identify the optimal temperature. Cheng et al. [17] introduced the stepwise refining TES (SRTES) method, which treats emissivity as a constant and the Planck function as a linear function of the wavenumber within a narrow spectral range (<15 cm^{-1}). SRTES uses the residue of the atmospheric downwelling radiance in the calculated ground object radiance spectrum as a cost function and employs a stepwise refining approach to determine both emissivity in the narrow spectral region of atmospheric emission lines and temperature. Meanwhile, Li et al. [18] proposed a correlation-wavelet method for TES, applying wavelet transforms to the ground-leaving radiance spectrum. In this method, the emission radiance spectrum of the ground object, derived using low-pass filtering, serves as the base frequency signal, while the atmospheric downward radiation is treated as the high-frequency signal. The optimal temperature is then determined by comparing the correlation coefficients between the emissivity spectrum calculated from the ground-leaving radiance spectrum and the emission radiance spectrum of the ground object at various input temperatures. The method also utilizes correlation to simultaneously estimate the proportion of wavelet signals at different scales to synthesize the emissivity spectrum, assuming that the atmospheric absorption peaks and valleys show similar deviations due to the discrepancies between the true values of the retrieved emissivity and the atmospheric downward spectrum. Lastly, Chen et al. [19] developed the temperature and emission retrieval from low-emissivity materials (TESLEM) method, designed for retrieving temperature and emissivity from low-emissivity ground objects.

Several studies have sought to reduce the number of unknowns in TES by expressing emissivity in a low-dimensional form. This method transforms the under-determined equations into over-determined ones, enabling successful retrieval of both emissivity and temperature. Three primary approaches have been explored: segmenting the emissivity spectrum, representing emissivity as a function of low-frequency wavelet coefficients, and assuming that the emissivity spectra of natural and man-made objects can be represented within a subspace of the original emissivity space. In the first approach, the emissivity spectrum is treated as a linear function of wavelength, where algorithms such as the linear spectral emissivity constraint (LSEC) [20], the improved LSEC (I-LSEC) [21], and the pre-estimated shape LSEC (PES-LSEC) [22] are employed. The second approach reduces the number of unknowns from $N + 1$ to $N/2 + 1$, utilizing the wavelet transform method for TES (WTTES) [23]. Finally, the third approach, known as the dictionary-based subspace TES (D-SBTES), proposed by Acito et al. [24], [25] and Ma et al. [26], represents the emissivity spectra within a predefined subspace, simplifying the inversion process.

From a spectral analysis perspective, most TES methods follow a similar approach with the exception of TISI. In general, the atmospheric absorption lines are much narrower than the spectral features of ground objects, allowing for the approximation of a degenerate spectral curve that closely resembles the true spectrum. The optimal temperature is determined through a cost function, and the corresponding emissivity is subsequently calculated using the radiative transfer equation.

Recent advancements in TES algorithms have been developed to enhance both temperature retrieval accuracy and emissivity estimation. For instance, Ma et al. [26] proposed a resolution-degrade-based spectral smoothness (RDSS) algorithm designed to simultaneously reduce temperature retrieval errors and improve the accuracy of temperature and emissivity. In this method, noise is further minimized through synchronized filtering of all variables in the radiative transfer equation, and the emissivity is calculated based on temperature using prefiltered retrieval variables. Similarly, Gao et al. [28] introduced an algorithm for HyperCam, utilizing the maximum–minimum relative emissivity difference (MMD), which establishes an empirical model between emissivity and its statistical properties derived from spectral libraries. Zhao et al. [29] proposed the statistical iterative spectrum smooth temperature/uncertainty separation (S-ISSTES), whose cost function is based on the propagation of first- and second-order statistics, aimed at supporting analysis, simulation, and system performance evaluation. Furthermore, machine learning has been increasingly applied to infrared hyperspectral emissivity inversion research [30].

Over the past 30 years, numerous TES algorithms have been proposed, demonstrating high retrieval accuracies. However, most of these algorithms have been tested primarily with simulation data within the framework of land resource monitoring, often favored over measured data due to the early-stage development of thermal infrared hyperspectral imagers, limited data availability, and concerns over data quality. Simulation data is also preferred for the convenience it provides in directly analyzing algorithm performance. While actual thermal infrared hyperspectral data introduces complexity due to the presence of various noise sources, retrieving ground object temperature and emissivity from such data remains a core objective of TES algorithms. Therefore, it is essential to expand the applicability of these algorithms and conduct more tests using real-world thermal infrared hyperspectral data on representative TES algorithms. Current data from thermal infrared hyperspectral systems offer a valuable opportunity to refine these algorithms. In this study, we developed a TES algorithm designed to accommodate a wider range of emissivity under minimal constraints. The thermal infrared hyperspectral data acquired by airborne thermal infrared hyperspectral imaging system (ATHIS) was utilized in the retrieval experiments.

II. Development of TES Algorithm

A. Basic Theory

In thermal infrared remote sensing, a simplified atmospheric radiative transfer equation is generally used to represent the

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radiance received by the sensor on the platform. The atmospheric radiative transfer equation in the thermal infrared region can be expressed as

$$L(\lambda_k) = L_g(\lambda_k)\tau\left(\lambda_k\right) + L_u(\lambda_k) \tag{1}$$

$$L_g(\lambda_k) = \varepsilon\left(\lambda_k\right)B(\lambda_k, T) + \left[1 - \varepsilon\left(\lambda_k\right)\right]L_d(\lambda_k) \tag{2}$$

where $L(\lambda_k)$ is the sensor radiance of band k; λ_k is the central wavelength of band k; $L_g(\lambda_k)$ is the ground-leaving radiance; $\varepsilon(\lambda_k)$ is the emissivity of the ground object; $L_d(\lambda_k)$ is the atmospheric downward radiance; $\tau(\lambda_k)$ is the transmittance of the atmosphere; $L_u(\lambda_k)$ is the path radiance of the atmosphere; $B(\lambda_k, T)$ is the Planck function; and T is the temperature of the ground object. The objective of temperature and emissivity retrieval is to solve the temperature and emissivity from the ground-leaving radiance equation [see (2)].

B. Improved Algorithm of TES

For the TES algorithm of the spectral smoothing iteration type, the spectral smoothing method and cost function were the two core components. Based on the principle, we developed a composite iterative spectral smoothing algorithm (CISSA). As the emissivity of a ground object decreases, the proportion of atmospheric downward radiation spectral features in the estimated emissivity spectrum increases. Consequently, these features are no longer represented as small spikes, particularly in the ozone absorption region. Therefore, the cost function was modified to enhance its suitability for TES applied to low-emissivity ground objects. To achieve this, the weight of the ozone absorption region was increased, and the composite cost function was upgraded as follows:

$$\sigma_{\mathrm{CISSA}}(\hat{T}) = \omega_1\sigma_1^*(\hat{T}) + \omega_2\sigma_2^*(\hat{T}) + \omega_3\sigma_3^*(\hat{T}) \tag{3}$$

where σ_1^* is the deviation of ground-leaving radiance; σ_2^* is the deviation of ground-leaving radiance in the spectral range of ozone absorption; $\hat{T}$ is the temperature value; σ_3^* is the degree to which emissivity exceeds the upper bound; ω_1, ω_2, and ω_3 are the weights of σ_1^*, σ_2^*, and σ_3^*, respectively.

In the cost function, the weight value of σ_2^* can be determined pixel by pixel based on the similarity between the ground-leaving radiance and the atmospheric downward radiance spectra. In this study, we use the correlation coefficient between the ground-leaving radiance spectrum and atmospheric downward radiance spectrum to determine ω_2. When the correlation coefficient is less than 0, ω_2 is 0. When the correlation coefficient is greater than or equal to 0, ω_2 is equal to the correlation coefficient. ω_1 and ω_3 are set to 0.5. To avoid the influence of dimensional discrepancies, each subcost function is standardized before calculating the weighted average. The calculation method for each subcost function in (3) is detailed in Table I.

In the context of spectral smoothing, reducing the influence of noise and singular values remains a pressing challenge. In CISSA, we adapted the method of RDSS [27] to minimize the effects of noise, and added a premedian filtering to remove outliers, which can improve the robustness of spectral smoothing. Fig. 1 shows the flowchart of retrieving temperature and emissivity. The step-by-step process of the improved TES algorithm is as follows.

TABLE I
CALCULATION FORMULAS OF COST FUNCTION

Function	Formula
$\sigma_1^*(\hat{T})$	$\sigma_1^*(\hat{T}) = \dfrac{\left[\sigma_1(\hat{T}) - \bar{\sigma}_1(\hat{T})\right]}{\bar{\sigma}_{1,SD}(\hat{T})}$
	$\sigma_1(\hat{T}) = \sqrt{\dfrac{1}{N_k}\sum_{k=1}^{N_k}\left[L_{g,fit}(\lambda_k, \hat{T}) - L_g(\lambda_k)\right]^2}$
	$L_{g,_{fit}}(\lambda_k) = \tilde{\varepsilon}(\lambda_k)B(\lambda_k, \hat{T}) + [1 - \tilde{\varepsilon}(\lambda_k)]L_d(\lambda_k)$
$\sigma_2^*(\hat{T})$	$\sigma_2^*(\hat{T}) = \dfrac{\left[\sigma_2(\hat{T}) - \bar{\sigma}_2(\hat{T})\right]}{\bar{\sigma}_{2,SD}(\hat{T})}$
	$\sigma_2(\hat{T}) = \sqrt{\dfrac{1}{B_2 - B_1 + 1}\sum_{k=B_1}^{B_2}\left[L_{g,fit}(\lambda_k, \hat{T}) - L_g(\lambda_k)\right]^2}$
	$L_{g,_{fit}}(\lambda_k) = \tilde{\varepsilon}(\lambda_k)B(\lambda_k, \hat{T}) + [1 - \tilde{\varepsilon}(\lambda_k)]L_d(\lambda_k)$
$\sigma_3^*(\hat{T})$	$\sigma_3^*(\hat{T}) = \dfrac{\left[\sigma_3(\hat{T}) - \bar{\sigma}_3(\hat{T})\right]}{\bar{\sigma}_{3,SD}(\hat{T})}$
	$\sigma_3(\hat{T}) = \sum_{j=1}^{N_k} sign\left[\hat{\varepsilon}(\lambda_j)\right]$
	$sign(x) = \begin{cases} 1, x > 1 \\ 0, x \le 1 \end{cases}$

Note: $\bar{\sigma}_X$ and $\bar{\sigma}_{X,SD}$ is the mean and standard deviation of σ_X (X = 1,2,3).

1) *Filtering Each Variable in the Ground-Leaving Radiative Transfer Equation:* Prior to calculating the emissivity, the average filtering was used to filter each variable. The calculation method was as follows:

$$\tilde{V} = F(V) \tag{4}$$

where F is the filtering function, V is the unfiltered variable, and $\tilde{V}$ is the filtered variable corresponding to V. In this study, the average filtering was adopted as the filtering function to filter the variables using the following calculation method below:

$$\left[\tilde{L}_g(\lambda_k), \tilde{L}_d(\lambda_k), \tilde{B}(\lambda_k, \hat{T})\right] = \frac{1}{2w_1 + 1}\sum_{j=-w_1}^{w_1}\left[L_g(\lambda_{k+j}), L_d(\lambda_{k+j}), B(\lambda_{k+j}, \hat{T})\right] \tag{5}$$

where $\tilde{L}_g(\lambda_k)$, $\tilde{L}_d(\lambda_k)$, and $\tilde{B}(\lambda_k, \hat{T})$ represent the filtered $L_g(\lambda_k)$, $L_d(\lambda_k)$, and $B(\lambda_k, \hat{T})$, respectively; w_1 are the radii of the filtering window.

2) *Calculation of Emissivity at a Given Temperature:* In the radiative transfer equation, the temperature and emissivity spectra exhibited a one-to-one correspondence. When the temperature was given, the corresponding emissivity spectrum can be obtained using the radiative transfer equation. In this study, the temperature was calculated

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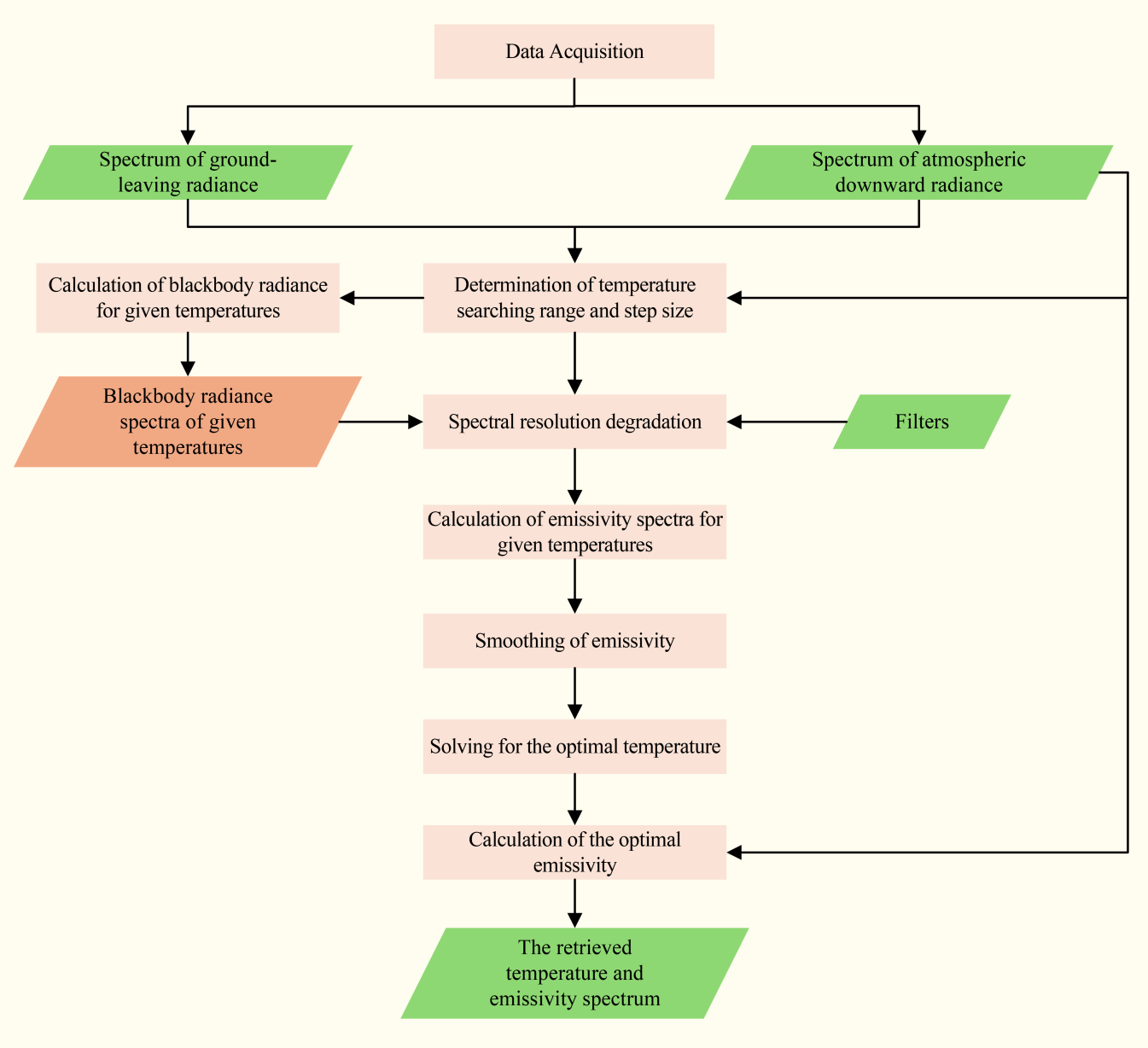


Fig. 1. Flowchart of retrieving temperature and emissivity.

as follows based on the variable filtered by (5):

$$\hat{\varepsilon}\left(\lambda_k, \hat{T}\right) = \frac{\tilde{L}_g\left(\lambda_k\right) - \tilde{L}_d\left(\lambda_k\right)}{\tilde{B}\left(\lambda_k, \hat{T}\right) - \tilde{L}_d\left(\lambda_k\right)} \tag{6}$$

where $\hat{\varepsilon}(\lambda_k)$ is the calculated emissivity spectrum.

3) *Smoothing of Emissivity:* The average filtering was adopted in the emissivity smoothing to remove burrs caused by atmospheric absorption and noise to allow the emissivity to be closer to the trend of the emissivity variation. The calculation method is as follows:

$$\varepsilon'\left(\lambda_k\right) = \text{median}\left[\hat{\varepsilon}\left(\lambda_j\right), w_2\right] \tag{7}$$

$$\tilde{\varepsilon}\left(\lambda_k\right) = \frac{1}{2w_2 + 1} \sum_{j=k-w_2}^{k+w_2} \varepsilon'\left(\lambda_j\right) \tag{8}$$

where $\tilde{\varepsilon}(\lambda_k)$ is the smoothed emissivity; and w_2 is the radius of the filtering window; median$(\cdot)$ is the median filtering function. The size of the filtering window determines whether the smoothing process can accurately de-spike and allow the smoothed emissivity closer to the actual value—if the filtering window was too large, the spikes were completely removed and the emissivity spectrum was distorted. However, when the filtering window was too small, residual spikes occur. Previous simulation experiments have shown that the retrieval error was minimal when the window size was set to three [13], [26]. Theoretically, this is related to the variation in atmospheric downward radiance, surface emissivity spectrum, and noise with wavelength. Hence, different pixels may have different optimal window sizes in a single thermal infrared hyperspectral data cube. For example, Shao et al. [27] showed that with an increase in noise in a size 3 window, the optimal temperatures gradually decreased. Contrastingly, the optimal temperatures

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gradually increased with an increased proportion of larger window sizes (5, 7, 9, etc.). Hence, when the data noise is large, appropriately increasing the window may contribute to maintaining retrieval accuracy.

4) *Solving for the Optimal Temperature:* The iterative solution to determine the optimal temperature was to minimize the cost function. Although there have been few search algorithms used to solve the optimal temperature value [13], [23], the inch-by-inch search was adopted to avoid local minima. The cost function was calculated according to (1), and the optimal value of temperature was determined by $\hat{T}_{\text{opt}}$.
5) *Calculation of the Optimal Emissivity:* The optimal estimated value of temperature was substituted into (9), and the final emissivity was calculated using the unfiltered variables. The calculation is given by the following equation:

$$\hat{\varepsilon}_{\text{opt}}(\lambda_i) = \frac{L_g(\lambda_i, T) - L_d(\lambda_i)}{B(\lambda_i, \hat{T}_{\text{opt}}) - L_d(\lambda_i)} \tag{9}$$

where $\hat{\varepsilon}_{\text{opt}}(\lambda_i)$ is the finally retrieved emissivity.

III. Data and Data Processing

Several typical TES algorithms including ISSTES, ARTEMISS, DRRI, TESLEM, WTTES, and RDSS were compared with the proposed algorithm (CISSA). The data cube acquired by ATHIS and the setup of algorithms were detailed in Sections III-A–III-C.

A. Data

The thermal infrared hyperspectral data cube used in this study was acquired using ATHIS during an aerial remote sensing experiment conducted on May 24, 2019, in Dongyang City, Zhejiang Province, China (120°18′11″E, 29°09′20″N) [30]. The relative altitude and ground spatial resolution were 1000 and 2 m, respectively. A synchronous ground measurement experiment was also performed, using a polytetrafluoroethylene (PTFE) film as a reference target. The weather was sunny during the experiment, with an air temperature of approximately 26.0 °C. The experimental site, located in a subtropical climate zone, features a complex landscape including mountainous terrain and diverse landforms such as buildings, roads, lakes, rivers, abandoned quarries, grasslands, and farmlands, providing an ideal setting for algorithm validation (Fig. 2). The data cube dimensions were 320 samples × 16 257 lines × 110 bands.

B. Data Preprocessing

The preprocessing of ATHIS data includes the following: abnormal pixel repair, radiometric calibration, nonuniformity correction, ground-based absolute radiometric calibration, and atmospheric correction. The implementation of these processes primarily followed the methods described by Liu et al. [30]. During preprocessing, the raw digital numbers were converted into physical radiance, which then served as input for retrieval. Fig. 3 shows false-color composite images of the ATHIS data before and after preprocessing, where the stripe noise was effectively removed. The ground objects were also more clearly defined in terms of outline and hierarchical structure.

TABLE II
Selected Bands of ATHIS for DRRI

	Left(μm)	Central(μm)	Right(μm)
1	10.233	10.262	10.291
2	10.522	10.551	10.580
3	10.811	10.840	10.869
4	10.985	11.014	11.043

TABLE III
Selected Bands of ATHIS for TESRLEM

	Peak(μm)	Valley(μm)
1	8.119	8.090
2	8.147	8.176
3	8.318	8.347
4	8.376	8.347
5	8.493	8.522
6	8.551	8.522
7	8.551	8.581
8	8.610	8.581
9	8.759	8.787
10	8.844	8.816

C. Settings of the TES Algorithms

1) Bands Selection of DRRI: According to Zhao et al. [29], the nominal spectral range of ATHIS is 8.5–12.5 μm. Due to the instrument's low sensitivity in the 11.2- to 12.5-μm range, the acquired thermal infrared hyperspectral data only covered 8.0–11.2 μm. As a result, it did not capture the distribution of the six selected band groups, which fall within the 11.2- to 12.5-μm range. Consequently, the DRRI methodology was deprioritized. Subsequently, four band groups were selected from the 8.0- to 11.2-μm range. One group from the ATHIS data was selected for DRRI, as its wavelengths were closer to the DRRI distribution (Table II).

2) Bands Selection of TESRLEM: In the TESRLEM algorithm, the bands selected by Chen et al. [19] were mainly distributed within the range of 8.0–8.9 μm. For the selected peaks and valleys of the downward radiance spectrum, the selected TESRLEM wavelengths were not entirely consistent with the central ATHIS wavelengths; leading to potential deviations of several to >10 μm. The bands closest to the wavelengths defined by Chen et al. [19] were selected in this study for operational convenience, as shown in Table III.

3) Decomposition Level of WTTES: Generally, higher wavelet decomposition levels result in more detailed displays and smoother postfiltered spectrums. Previous simulation experiments showed that the selection of wavelet bases had minimal effect on the algorithm, while the wavelet decomposition level itself had a relatively larger effect [23]. Accordingly, the third-level wavelet decomposition was found

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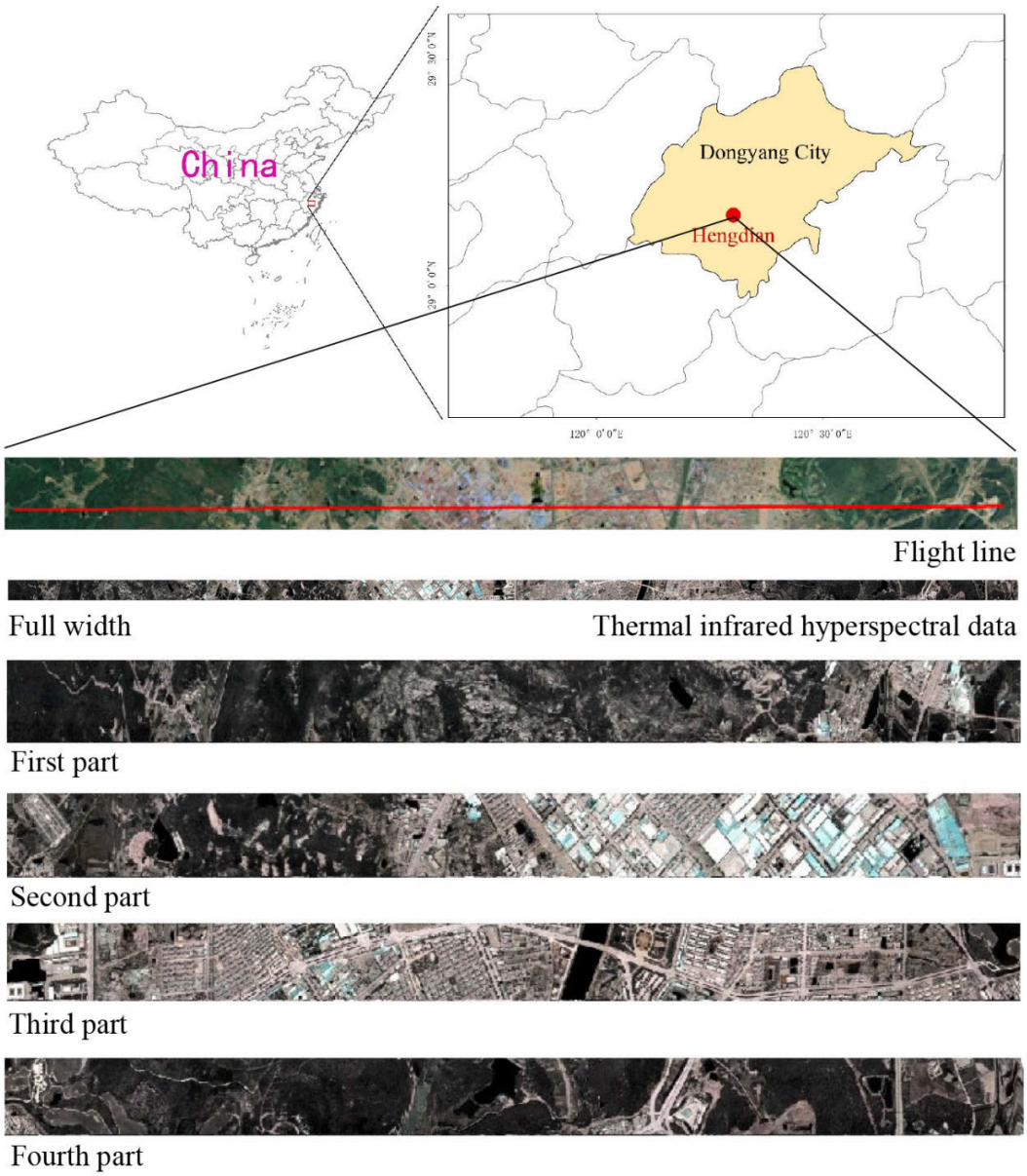


Fig. 2. Schematic map of the flight line and data cube. In order to show the details of the thermal infrared hyperspectral data, the entire image is enlarged four times and divided into four parts for display.

to be the most effective. Therefore, the Daubechies wavelet function was considered as the wavelet base in this study. The decomposition level was also set to levels two, three, and five to allow for a more comprehensive investigation on the performance of WTTES. Consequently, these were assigned as WTTES-LV2, WTTES-LV3, and WTTES-LV5, respectively.

4) Determination of Optimal Temperature: As the temperatures of ground objects are typically considered to be within a certain range, an inch-by-inch search was adopted to determine the optimal temperature and avoid local minima. This involved setting the temperature searching range $[T_{\text{inf}}, T_{\text{sup}}]$ and step size (δT) based on prior knowledge. Thereafter, the candidate temperature set was established, and the cost function corresponding to all temperatures in the candidate temperature set was calculated, in which the temperature mapped to the minimum of the cost function was considered the optimal temperature. In this study, the temperature search range and step size were [0, 100] °C and 0.1 °C, respectively,

$$T_{\text{opt}} = \min_{T \in [T_{\text{inf}}, T_{\text{sup}}]} \operatorname{argmin}(\sigma). \tag{10}$$

IV. Results

A. Results of Temperature Retrieval

Fig. 4 shows the temperature histograms retrieved by the different algorithms. Figs. 5 and 6 present the downtown temperature and emissivity images, respectively, which highlight the areas excluded from the temperature and emissivity data retrieved from the ATHIS data cube. Table IV lists the statistical parameters of the histograms in Fig. 3. Based on the visual characteristics, the retrieved temperature images can be classified into three groups based on algorithm retrieval: those retrieved by: 1) ISSTES; 2) DRRI and TESLEMI; and

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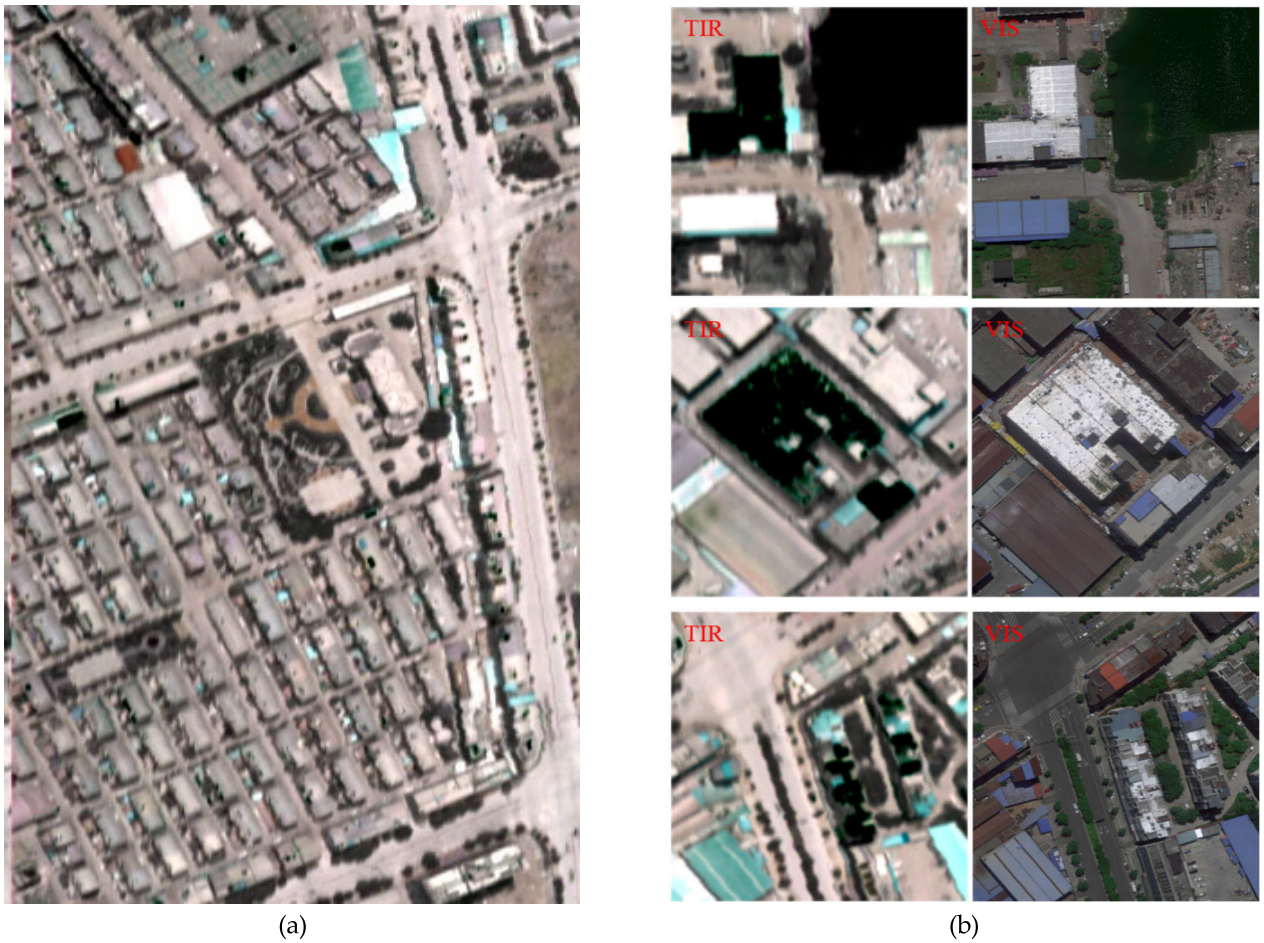


Fig. 3. False color composite images of ATHIS data. (a) False color composite images of downtown (R: 10.644 μm, G: 9.485 μm, B: 8.906 μm). (b) Examples of low-emissivity ground objects. TIR represents the color composite image of the ATHIS data (R: 10.644 μm, G: 9.485 μm, B: 8.906 μm). Visible (VIS) represents the true color composite images that was synchronously acquired. The spatial resolution of TIR images is 1 m, and the VIS is 0.25 m.

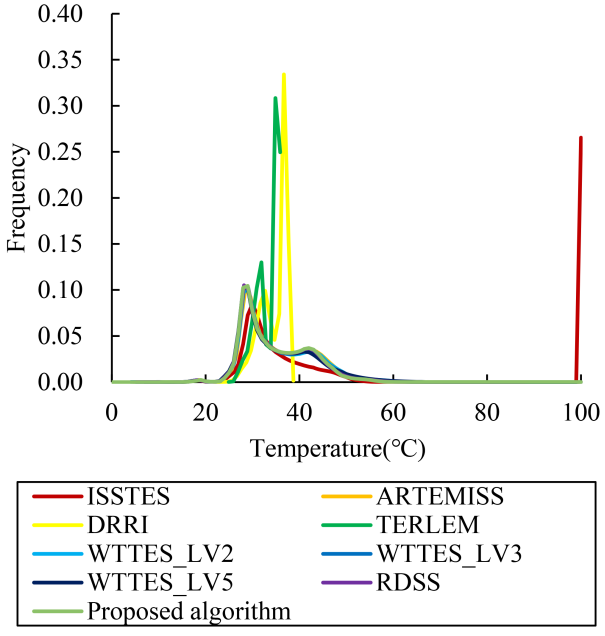


Fig. 4. Temperature histograms of different TES algorithms.

3) ARTEMISS, WTTES_LV2, WTTES_LV3, WTTES_LV5, RDSS, and CISSA.

For the first group, the outlines of ground objects were basically visible; however, numerous pixels with incorrect temperature values can be noted. As shown in the temperature histogram and statistical parameters, most of the incorrect values were equivalent to 100, indicating the failure of obtaining optimal temperatures for the pixels.

The second group reveals significant distortion in the retrieved temperatures, where the outlines of ground objects are not clearly visible. This distortion may be attributed to the highly concentrated temperature values produced by the two algorithms. However, this does not necessarily indicate a significant error. Rather, it may suggest that the compromise band selections described in Section III-C are not suitable. Additionally, these two methods contribute to TES only when specific requirements for the instrument's band setup are met.

In the third group, the outlines of ground objects in the temperature images were clearly defined. The histograms were bimodal distributions of one large peak and one small peak, which were consistent with the actual temperature distribution. Accordingly, high-temperature peaks accurately corresponded to artificial buildings, bare land, and rocks, whereas low-temperature peaks corresponded to vegetation, water, and crops.

The comparison between the temperature and emissivity images showed that the TES process can also be considered

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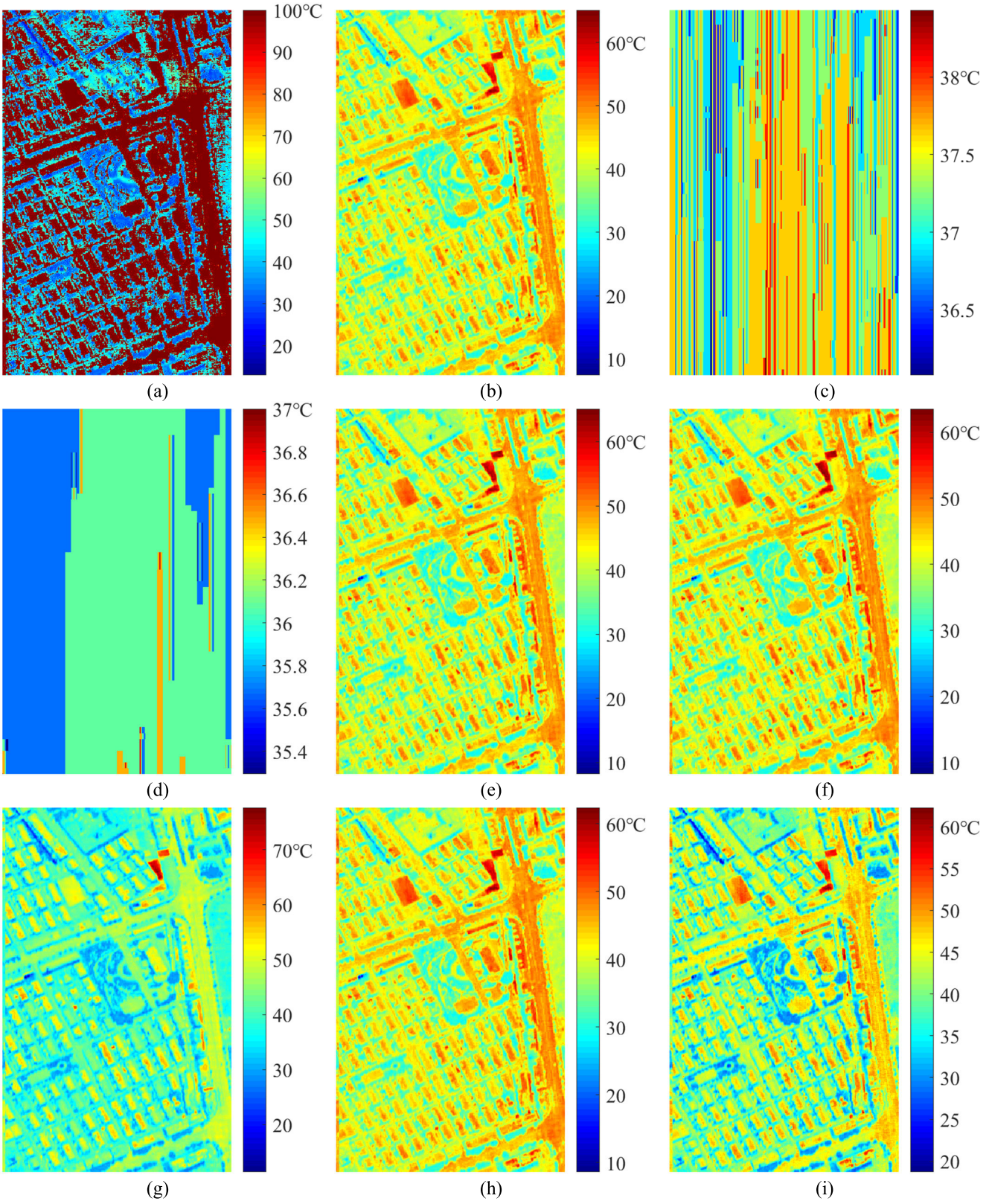


Fig. 5. Downtown temperature images retrieved by the algorithms from the ATHIS data cube. (a) ISSTES, (b) ARTEMISS, (c) DRRI, (d) TESLEMI, (e) WTTES_LV2, (f) WTTES_LV3, (g) WTTES_LV5, (h) RDSS, and (i) proposed algorithm.

as an information distribution medium—when temperature retrieval was distorted, the temperature information was consequently fused into the emissivity image, causing emissivity distortion. However, when the temperature retrieval was accurate, the emissivity was also more accurate.

B. Results of Emissivity Retrieval

Figs. 7 and 8 show the emissivity spectra of the rocks in a quarry and PTEF target retrieved by different algorithms, respectively. The measured emissivity spectra were acquired using a model 102F Fourier-transform infrared spectrometer.

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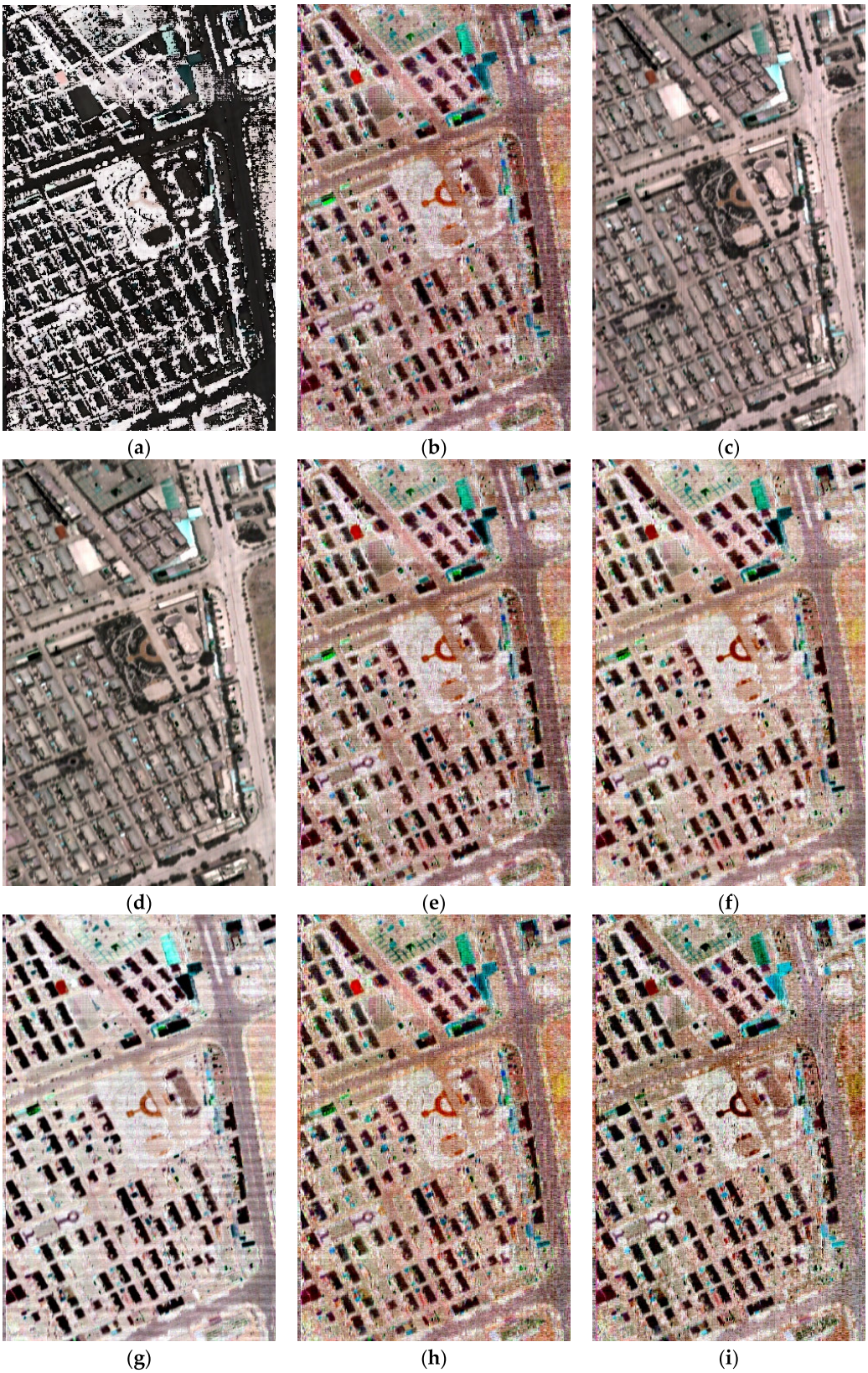


Fig. 6. False color composite images of downtown emissivity retrieved by different algorithms from the ATHIS data cube. R: 10.644 μm, G: 9.485 μm, B: 8.906 μm. (a) ISSTES, (b) ARTEMISS, (c) DRRI, (d) TESLEMI, (e) WTTES_LV2, (f) WTTES_LV3, (g) WTTES_LV5, (h) RDSS, and (i) proposed algorithm.

It can be observed that the retrieved emissivity spectra were significantly similar to the measured spectra, in which the primary difference was the distinct emissivity values of the algorithms. The peaks and valleys of the measured spectra were also consistent with those of the retrieved spectra. For instance, the spectral characteristics of the rocks were

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TABLE IV
STATISTICAL PARAMETERS OF TEMPERATURE HISTOGRAMS OF THE TES ALGORITHMS

Algorithm	Minimum(◎)	Maximum(◎)	Mean(◎)	Standard deviation(◎)
ISSTES	15.0	100.0	52.1	29.3
ARTEMISS	4.6	100.0	35.8	7.4
DRRI	21.7	39.0	35.1	2.8
TESLEMI	24.9	36.8	33.9	2.4
WTTES_LV2	5.1	100.0	35.7	7.5
WTTES_LV3	4.1	100.0	35.5	7.3
WTTES_LV5	7.0	100.0	35.5	7.6
RDSS	5.2	89.0	35.3	7.1
Proposed Algorithm	12.8	83.6	35.5	7.2

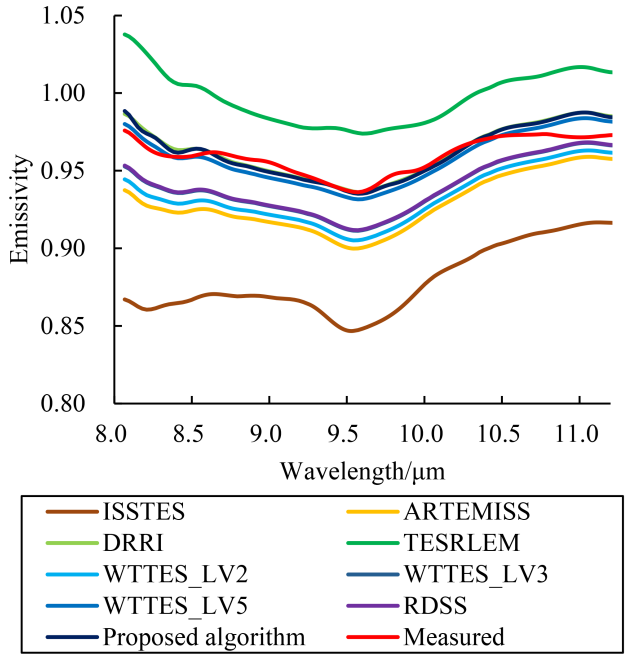


Fig. 7. Emissivity spectra of rocks in the quarry retrieved by different algorithms.

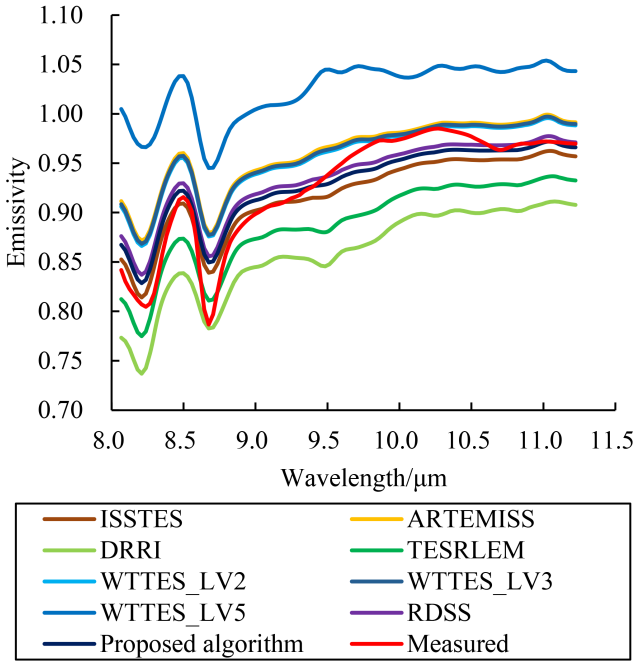


Fig. 8. Emissivity spectra of the target retrieved by different algorithms.

9.57 μm, in which the targets were 8.24, 8.50, and 8.67 μm. Table V validates this inference, as the cosine values of the spectral angle between the retrieved spectra and measured spectra were all above 0.999. The absolute derivations of the different algorithms were quite different as the retrieval error of WTTES_LV5 was the lowest, with a value of 0.007 for the emissivity spectrum of the rock in the quarry, while the retrieval error of the proposed algorithm was the lowest, with a value of 0.014 for the emissivity spectrum of the target (Table VI). Generally, the proposed algorithm has the lowest average error compared to the others.

C. Retrieval of the Ground Objects With Low Emissivity

Typical ground objects with low emissivity mainly include rocks and minerals with strong reflection, metal targets, and targets with metal surfaces. In this study, several ground objects were found to have low emissivity in the data cube acquired by ATHIS. Fig. 2(b) shows three roofs covered with low-emissivity materials. The primary characteristics of the low-emissivity ground objects were their low emission and large reflected atmospheric downward radiation proportion; therefore, these would always appear dark in the color composite image.

Figs. 9–11 show the retrieved emissivity and temperature images, and average emissivity spectra of the first ground object in Fig. 2(b), respectively. Generally, the retrieved results for the proposed algorithm were more accurate than those for ARTEMISS, WTTES_LV2, WTTES_LV3, WTTES_LV5, and RDSS. The average temperature of the proposed algorithm also obtained the highest with 18.8 °C (Table VII).

The low emissivity retrieved by the proposed algorithm was further reduced, and the spectral feature of atmospheric downward radiation in the ozone absorption region was removed. The retrieved temperatures of the pixels belonging to the first

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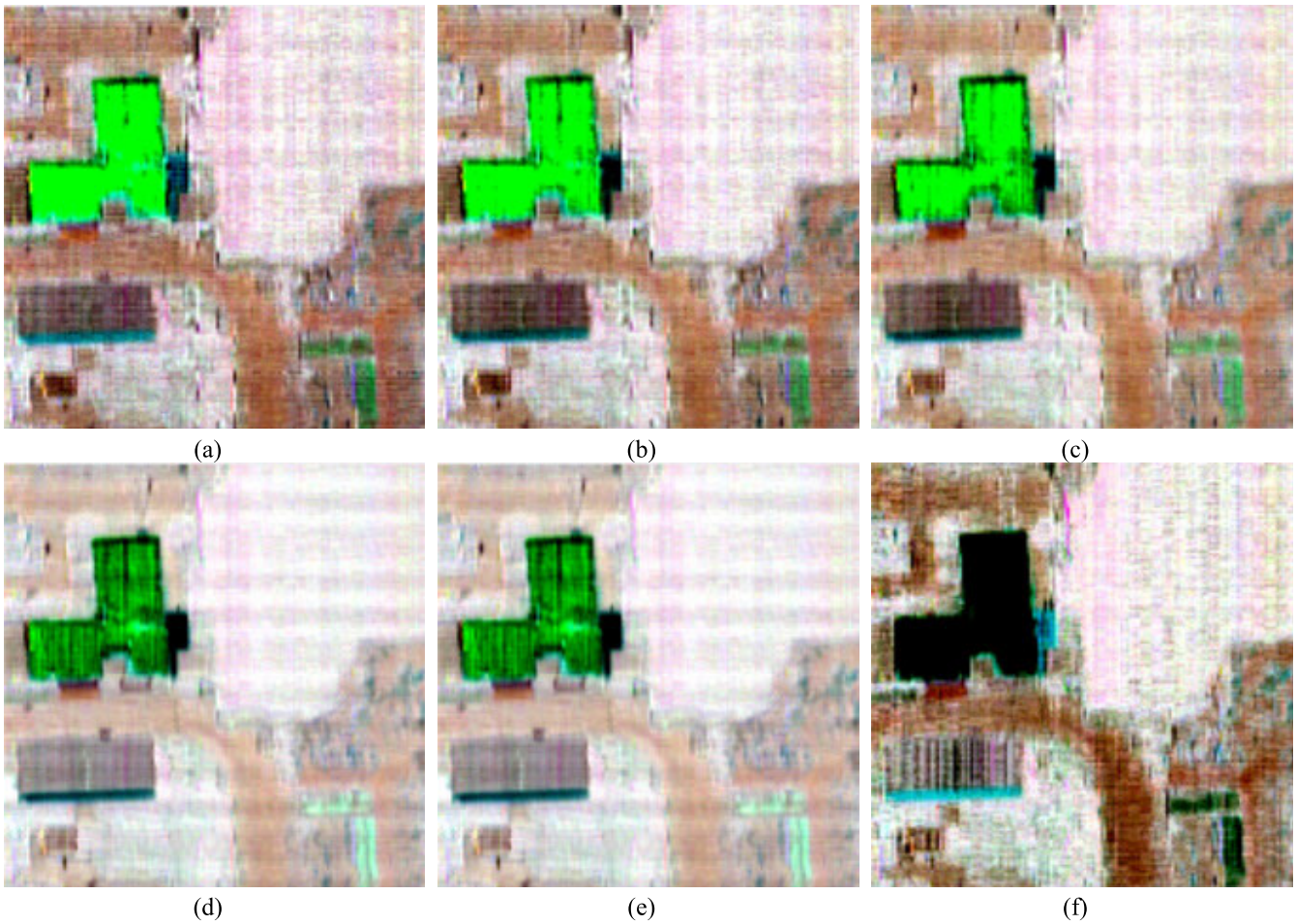


Fig. 9. False color composite images of the retrieved emissivity for the ground objects with low emissivity. R: 10.644 μm, G: 9.485 μm, B: 8.906 μm. (a) ARTEMISS, (b) WTTES_LV2, (c) WTTES_LV3, (d) WTTES_LV5, (e) RDSS, and (f) proposed algorithm.

TABLE V
SPECTRAL ANGLE COSINES BETWEEN THE MEASURED AND RETRIEVED SPECTRA

	Rock in the quarry	Target of PTEF	Mean
ISSTES	1.000	1.000	1.000
ARTEMISS	1.000	1.000	1.000
DRRI	1.000	1.000	1.000
TESRLEM	1.000	1.000	1.000
WTTES_LV2	1.000	1.000	1.000
WTTES_LV3	1.000	1.000	1.000
WTTES_LV5	1.000	0.999	1.000
RDSS	1.000	1.000	1.000
Proposed algorithm	1.000	1.000	1.000

TABLE VI
ABSOLUTE DERIVATIONS BETWEEN THE MEASURED AND RETRIEVED SPECTRA

	Rock in the quarry	Target of PTEF	Mean
ISSTES	0.082	0.022	0.052
ARTEMISS	0.033	0.040	0.037
DRRI	0.011	0.071	0.041
TESRLEM	0.040	0.043	0.042
WTTES_LV2	0.028	0.036	0.032
WTTES_LV3	0.023	0.038	0.031
WTTES_LV5	0.010	0.102	0.056
RDSS	0.019	0.021	0.020
Proposed algorithm	0.007	0.020	0.014

ground object with low emissivity were significantly improved, in which the average value was 24.6 °C. This retrieved value was consistent with the actual conditions as it was relatively close to the actual air temperature. Compared with other algorithms, the increase in the maximal temperature retrieved by the proposed algorithm was only 1.2 °C, which was significantly less than the increase in the average temperature (5.8 °C). This indicates that the retrieved temperature of pixels belonging to the ground objects with non-low-emissivity in Fig. 2(b) obtained by the proposed algorithm was not an overall improvement; however, a more effective improvement on the temperatures of pixels with low emissivity.

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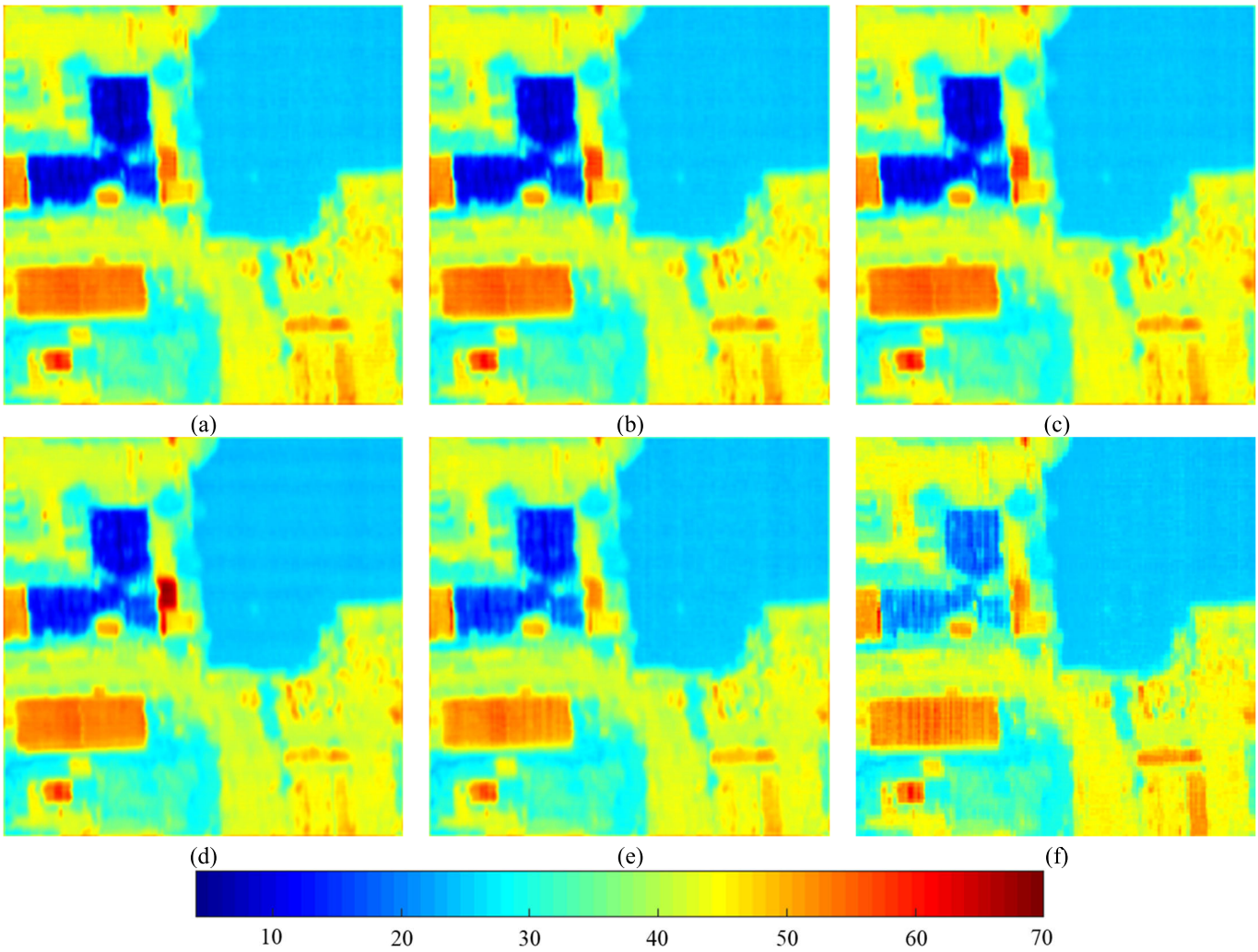


Fig. 10. Retrieved temperature images of the ground object with low emissivity. (a) ARTEMISS, (b) WTTES_LV2, (c) WTTES_LV3, (d) WTTES_LV5, (e) RDSS, and (f) proposed algorithm.

TABLE VII
MEANS OF THE RETRIEVED TEMPERATURES OF THE GROUND OBJECT WITH LOW EMISSIVITY

Method	MIN(℃)	MEAN(℃)	MAX(℃)
ARTEMISS	4.6	13.2	35.9
WTTES_LV2	5.1	14.0	36.3
WTTES_LV3	4.1	14.5	37.4
WTTES_LV5	7	16.6	38.4
RDSS	8.4	18.8	40.1
Proposed algorithm	15.2	24.6	41.3

D. Influence of Parameter Settings on Retrieval Errors of Proposed Algorithm

Shao et al. [27] have shown that when the thermal infrared hyperspectral data had a high noise level, increasing the window size of the variable filtering and emissivity smoothing can aid in improving the retrieval accuracy of the temperature emissivity. In the retrieval experiment of the proposed algorithm, 11 variable filtering windows and emissivity smoothing windows were tested, in which the window size was increased from 3 to 23 in steps of 2; 121 retrieval instances were obtained in total. Figs. 12 and 13 show the histograms and their statistics of the retrieved temperatures using different window size settings, in which variation trends of different window settings were found to be basically similar, except for minor differences in the local temperature values. This indicates that different window settings result in significant changes in the maximum and minimum of the retrieval results, whereas the mean and standard deviation have minimal effects.

Table VIII lists the retrieved emissivity errors compared with the measured emissivity spectra of the rock in a quarry, including the target under different window size settings. It can be observed that under different window settings, the emissivity error had significant variations, in which, a minimum value of 0.014 increased to a maximum value of 0.025 instantaneously. Fig. 14 shows the variation trend of emissivity retrieval error with window settings, in which the emissivity retrieval error initially increases and then decreases with the variable filtering window size (w_1). The window size with the lowest error was also notably located from 13 to 17. However,

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TABLE VIII
RETRIEVAL ERRORS OF EMISSIVITY UNDER DIFFERENT WINDOW SIZE SETTINGS

w_2 \ w_1	3	5	7	9	11	13	15	17	19	21	23
3	0.021	0.020	0.020	0.022	0.019	0.016	0.023	0.021	0.025	0.024	0.019
5	0.020	0.020	0.022	0.021	0.019	0.015	0.016	0.017	0.016	0.022	0.019
7	0.020	0.019	0.020	0.021	0.018	0.014	0.015	0.016	0.015	0.020	0.021
9	0.019	0.018	0.018	0.020	0.018	0.015	0.015	0.015	0.016	0.020	0.021
11	0.019	0.018	0.018	0.019	0.017	0.015	0.015	0.016	0.017	0.019	0.021
13	0.019	0.018	0.017	0.019	0.017	0.016	0.015	0.016	0.017	0.020	0.021
15	0.018	0.017	0.017	0.019	0.017	0.016	0.016	0.017	0.018	0.020	0.021
17	0.018	0.018	0.018	0.019	0.018	0.017	0.016	0.017	0.018	0.019	0.021
19	0.018	0.018	0.019	0.019	0.018	0.017	0.017	0.018	0.019	0.019	0.020
21	0.018	0.019	0.020	0.019	0.018	0.018	0.017	0.018	0.019	0.019	0.020
23	0.019	0.020	0.020	0.019	0.018	0.018	0.018	0.018	0.019	0.020	0.020

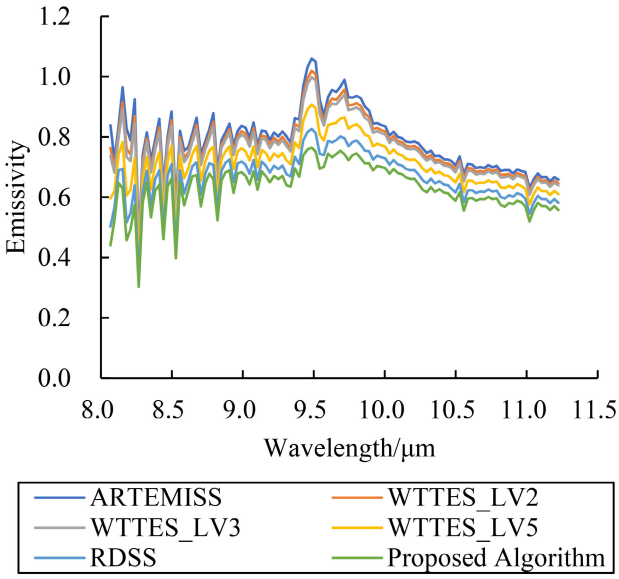


Fig. 11. Average emissivity spectra retrieved by different algorithms of the ground object with low emissivity.

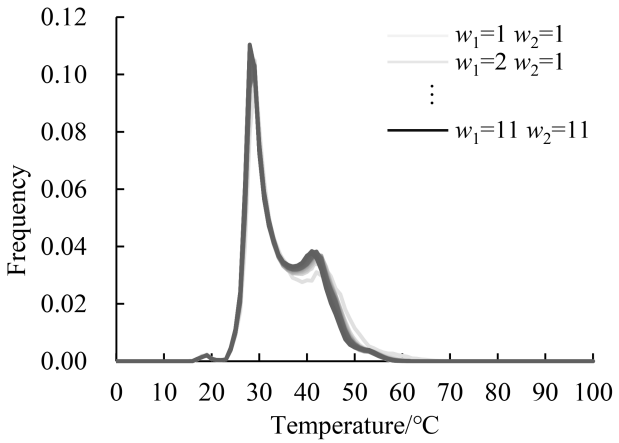


Fig. 12. Histograms of the retrieved temperatures using different window size settings.

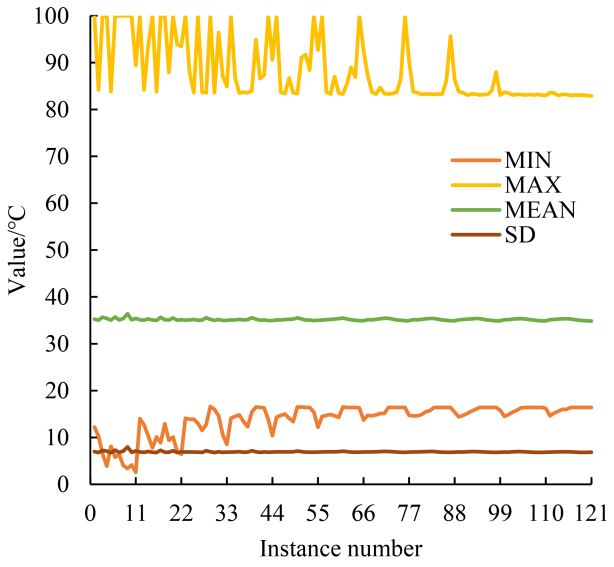


Fig. 13. Histogram statistics of retrieved temperatures using different window size settings.

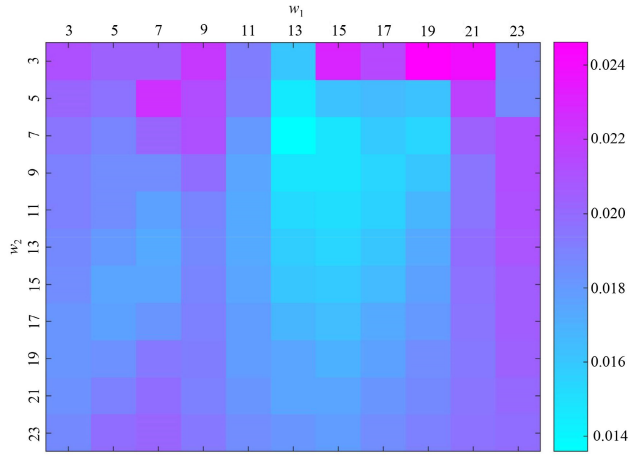


Fig. 14. Variation trend of retrieval errors with different settings of window size.

no apparent trend with the variable filtering window size (w_2) can be noted on the emissivity retrieval error. As shown in Table VIII, when the window size setting was found in the region of w_1 = 7–13 and w_1 = 13–17, the proposed

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algorithm achieves a lower retrieval error with 0.14–0.16. Therefore, when the proposed algorithm is used for the TES of data cube acquired by different thermal infrared hyperspectral instruments, setting the specific window size for various data quality is necessary.

V. Conclusion

To improve the retrieval quality of temperature and emissivity in the context of remote sensing with complex underlying surfaces using thermal infrared hyperspectral data, we developed the TES algorithm called CISSA, which is capable of adapting to a wider range of emissivity. Comparative experiments were conducted using the data cube acquired by ATHIS. Following algorithm improvements, data preprocessing, retrieval, and analysis, the following conclusions were drawn.

1) *Impact of Cost Function:* The cost function plays a significant and crucial role in the retrieval of temperature and emissivity. Differences were observed in the retrieved temperature histograms of ISSTES, DRRI, and TESLEMI, due to similarities in the cost functions and temperature histograms between ARTEMISS, WTTES_LV2, WTTES_LV3, WTTES_LV5, RDSS, and the proposed CISSA.
2) *Algorithm Performance:* Validation experiments demonstrated that CISSA slightly outperforms other mainstream algorithms for general surface coverage. However, for low-emissivity land cover, CISSA significantly improves TES accuracy, indicating its potential to adapt to more complex emissivity distributions. To fully solve this problem, it is also necessary to address the complex adjacent effects caused by the radiative transmission of 3-D structures.
3) *Effect of Algorithm Parameters:* The parameters of the TES algorithm have a substantial impact on the results. Optimal performance for ATHIS data was achieved when the window size was within a specific range. The optimal parameter settings were found to be closely related to the noise level and correction errors in the input data, making it essential to prioritize the adaptability of parameter settings. This issue is not exclusive to TES algorithms but is common in many remote sensing models.

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